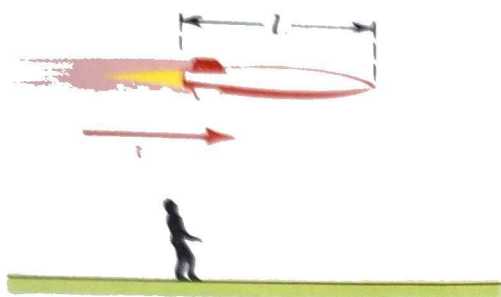


FIGURE 2.24



EXAMPLE 10 Let c denote the speed of light (approximately 3.0×10^8 m/sec, or 186,000 mi/sec). In Einstein's theory of relativity, the Lorentz contraction formula

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

specifies the relationship between (1) the length L of an object that is moving at a velocity v with respect to an observer and (2) its length L_0 at rest (see Figure 2.24). The formula implies that the length of the object measured by the observer is shorter when it is moving than when it is at rest. Find and interpret $\lim_{v \rightarrow c^-} L$, and explain why a left-hand limit is necessary.

SOLUTION Using (one-sided) limit theorems yields

$$\begin{aligned} \lim_{v \rightarrow c^-} L &= \lim_{v \rightarrow c^-} L_0 \sqrt{1 - \frac{v^2}{c^2}} \\ &= L_0 \lim_{v \rightarrow c^-} \sqrt{1 - \frac{v^2}{c^2}} \\ &= L_0 \sqrt{\lim_{v \rightarrow c^-} \left(1 - \frac{v^2}{c^2}\right)} \\ &= L_0 \sqrt{0} = 0. \end{aligned}$$

Thus, if the velocity of an object could approach the speed of light, then its length, as measured by an observer at rest, would approach zero. This result is sometimes used to help justify the theory that the speed of light is the ultimate speed in the universe; that is, no object can have a velocity that is greater than or equal to c .

A left-hand limit is necessary because if $v > c$, then $\sqrt{1 - (v^2/c^2)}$ is not a real number.

EXERCISES 2.3

Exer. 1–48: Use theorems on limits to find the limit, if it exists.

1 $\lim_{x \rightarrow \sqrt{2}} 15$

3 $\lim_{x \rightarrow -2} x$

5 $\lim_{x \rightarrow 4} (3x - 4)$

7 $\lim_{x \rightarrow -2} \frac{x - 5}{4x + 3}$

9 $\lim_{x \rightarrow 1} (-2x + 5)^4$

11 $\lim_{x \rightarrow 3} (3x - 9)^{100}$

13 $\lim_{x \rightarrow -2} (3x^3 - 2x + 7)$

2 $\lim_{x \rightarrow 15} \sqrt{2}$

4 $\lim_{x \rightarrow 3} x$

6 $\lim_{x \rightarrow -2} (-3x + 1)$

8 $\lim_{x \rightarrow 4} \frac{2x - 1}{3x + 1}$

10 $\lim_{x \rightarrow -2} (3x - 1)^5$

12 $\lim_{x \rightarrow 1/2} (4x - 1)^{50}$

14 $\lim_{x \rightarrow 4} (5x^2 - 9x - 8)$

15 $\lim_{x \rightarrow \sqrt{2}} (x^2 + 3)(x - 4)$

17 $\lim_{x \rightarrow \pi} (x - 3.1416)$

19 $\lim_{s \rightarrow 4} \frac{6s - 1}{2s - 9}$

21 $\lim_{x \rightarrow 1/2} \frac{2x^2 + 5x - 3}{6x^2 - 7x + 2}$

23 $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{(x - 2)^2}$

25 $\lim_{x \rightarrow -2} \frac{x^3 + 8}{x^4 - 16}$

27 $\lim_{x \rightarrow 2} \frac{(1/x) - (1/2)}{x - 2}$

16 $\lim_{t \rightarrow -3} (3t + 4)(7t - 9)$

18 $\lim_{x \rightarrow \pi} (\frac{1}{2}x - \frac{1}{7})$

20 $\lim_{x \rightarrow 1/2} \frac{4x^2 - 6x + 3}{16x^3 + 8x - 7}$

22 $\lim_{x \rightarrow 2} \frac{x - 2}{x^3 - 8}$

24 $\lim_{x \rightarrow -2} \frac{x^2 + 2x - 3}{x^2 + 5x + 6}$

26 $\lim_{x \rightarrow 16} \frac{x - 16}{\sqrt{x} - 4}$

28 $\lim_{x \rightarrow -3} \frac{x + 3}{(1/x) + (1/3)}$

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$$29 \lim_{x \rightarrow 1} \left(\frac{x^2}{x-1} - \frac{1}{x-1} \right)$$

$$30 \lim_{x \rightarrow 1} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^6$$

$$31 \lim_{x \rightarrow 16} \frac{2\sqrt{x} + x^{3/2}}{\sqrt[4]{x} + 5}$$

$$32 \lim_{x \rightarrow -8} \frac{16x^{2/3}}{4 - x^{4/3}}$$

$$33 \lim_{x \rightarrow 4} \sqrt[3]{x^2 - 5x - 4}$$

$$34 \lim_{x \rightarrow -2} \sqrt{x^4 - 4x + 1}$$

$$35 \lim_{x \rightarrow 3} \sqrt[3]{\frac{2 + 5x - 3x^3}{x^2 - 1}}$$

$$36 \lim_{x \rightarrow \pi} \sqrt[5]{\frac{x - \pi}{x + \pi}}$$

$$37 \lim_{h \rightarrow 0} \frac{4 - \sqrt{16 + h}}{h}$$

$$38 \lim_{h \rightarrow 0} \left(\frac{1}{h} \right) \left(\frac{1}{\sqrt{1+h}} - 1 \right)$$

$$39 \lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^5 - 1}$$

$$40 \lim_{x \rightarrow 2} \frac{x^2 - 7x + 10}{x^6 - 64}$$

$$41 \lim_{v \rightarrow 3} v^2(3v - 4)(9 - v^3)$$

$$42 \lim_{k \rightarrow 2} \sqrt{3k^2 + 4} \sqrt[3]{3k + 2}$$

$$43 \lim_{x \rightarrow 5^+} (\sqrt{x^2 - 25} + 3)$$

$$44 \lim_{x \rightarrow 3^-} x\sqrt{9 - x^2}$$

$$45 \lim_{x \rightarrow 3^+} \frac{\sqrt{(x-3)^2}}{x-3}$$

$$46 \lim_{x \rightarrow -10^-} \frac{x+10}{\sqrt{(x+10)^2}}$$

$$47 \lim_{x \rightarrow 5^-} \frac{1 + \sqrt{2x - 10}}{x + 3}$$

$$48 \lim_{x \rightarrow 4^+} \frac{\sqrt[4]{x^2 - 16}}{x + 4}$$

$$62 \lim_{x \rightarrow 0} \frac{|x|}{\sqrt{x^4 + 4x^2 + 7}} = 0$$

(Hint: Use $f(x) = 0$ and $g(x) = |x|$.)

$$63 \lim_{x \rightarrow 0} x \sin(1/x) = 0$$

(Hint: Use $f(x) = -|x|$ and $g(x) = |x|$.)

$$64 \lim_{x \rightarrow 0} x^4 \sin(1/\sqrt[3]{x}) = 0 \quad (\text{Hint: See Example 8.})$$

$$65 \text{ If } 0 \leq f(x) \leq c \text{ for some real number } c, \text{ prove that } \lim_{x \rightarrow 0} x^2 f(x) = 0.$$

$$66 \text{ If } \lim_{x \rightarrow a} f(x) = L \neq 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0, \text{ prove that } \lim_{x \rightarrow a} [f(x)/g(x)] \text{ does not exist. (Hint: Assume there is a number } M \text{ such that } \lim_{x \rightarrow a} [f(x)/g(x)] = M \text{ and consider } \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [g(x) \cdot f(x)/g(x)].)$$

$$67 \text{ Explain why } \lim_{x \rightarrow 0} \left(x \sin \frac{1}{x} \right) \neq \left(\lim_{x \rightarrow 0} x \right) \left(\lim_{x \rightarrow 0} \sin \frac{1}{x} \right).$$

$$68 \text{ Explain why } \lim_{x \rightarrow 0} \left(\frac{1}{x} + x \right) \neq \lim_{x \rightarrow 0} \frac{1}{x} + \lim_{x \rightarrow 0} x.$$

$$69 \text{ Charles' law for gases states that if the pressure remains constant, then the relationship between the volume } V \text{ that a gas occupies and its temperature } T \text{ (in } ^\circ\text{C)} \text{ is given by } V = V_0(1 + \frac{1}{273}T). \text{ The temperature } T = -273^\circ\text{C} \text{ is absolute zero.}$$

$$(a) \text{ Find } \lim_{T \rightarrow -273^+} V.$$

$$(b) \text{ Why is a right-hand limit necessary?}$$

$$70 \text{ According to the theory of relativity, the length of an object depends on its velocity } v \text{ (see Example 10). Einstein also proved that the mass } m \text{ of an object is related to } v \text{ by the formula}$$

$$m = \frac{m_0}{\sqrt{1 - (v^2/c^2)}},$$

where m_0 is the mass of the object at rest.

$$(a) \text{ Investigate } \lim_{v \rightarrow c^-} m.$$

$$(b) \text{ Why is a left-hand limit necessary?}$$

$$71 \text{ A convex lens has focal length } f \text{ centimeters. If an object is placed a distance } p \text{ centimeters from the lens, then the distance } q \text{ centimeters of the image from the lens is related to } p \text{ and } f \text{ by the lens equation}$$

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}.$$

As shown in the figure on the following page, p must be greater than f for the rays to converge.

$$(a) \text{ Investigate } \lim_{p \rightarrow f^+} q.$$

$$(b) \text{ What is happening to the image as } p \rightarrow f^+?$$

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$$29 \lim_{x \rightarrow 1} \left(\frac{x^2}{x-1} - \frac{1}{x-1} \right)$$

$$30 \lim_{x \rightarrow 1} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^6$$

$$31 \lim_{x \rightarrow 16} \frac{2\sqrt{x} + x^{3/2}}{\sqrt[4]{x} + 5}$$

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$$66 \text{ If } \lim_{x \rightarrow a} f(x) = L \neq 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0, \text{ prove that } \lim_{x \rightarrow a} [f(x)/g(x)] \text{ does not exist. (Hint: Assume there is a number } M \text{ such that } \lim_{x \rightarrow a} [f(x)/g(x)] = M \text{ and consider } \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [g(x) \cdot f(x)/g(x)].)$$

$$67 \text{ Explain why } \lim_{x \rightarrow 0} \left(x \sin \frac{1}{x} \right) \neq \left(\lim_{x \rightarrow 0} x \right) \left(\lim_{x \rightarrow 0} \sin \frac{1}{x} \right).$$

$$68 \text{ Explain why } \lim_{x \rightarrow 0} \left(\frac{1}{x} + x \right) \neq \lim_{x \rightarrow 0} \frac{1}{x} + \lim_{x \rightarrow 0} x.$$

$$69 \text{ Charles' law for gases states that if the pressure remains constant, then the relationship between the volume } V \text{ that a gas occupies and its temperature } T \text{ (in } ^\circ\text{C)} \text{ is given by } V = V_0(1 + \frac{1}{273}T). \text{ The temperature } T = -273^\circ\text{C} \text{ is absolute zero.}$$

$$(a) \text{ Find } \lim_{T \rightarrow -273^+} V.$$

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$$m = \frac{m_0}{\sqrt{1 - (v^2/c^2)}},$$

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$$(a) \text{ Investigate } \lim_{v \rightarrow c^-} m.$$

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$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f}.$$

As shown in the figure on the following page, p must be greater than f for the rays to converge.

$$(a) \text{ Investigate } \lim_{p \rightarrow f^+} q.$$

$$(b) \text{ What is happening to the image as } p \rightarrow f^+?$$

Exer. 49–52: Find each limit, if it exists:

$$(a) \lim_{x \rightarrow a^-} f(x) \quad (b) \lim_{x \rightarrow a^+} f(x) \quad (c) \lim_{x \rightarrow a} f(x)$$

$$49 f(x) = \sqrt{5-x}; \quad a = 5$$

$$50 f(x) = \sqrt{8-x^3}; \quad a = 2$$

$$51 f(x) = \sqrt[3]{x^3-1}; \quad a = 1$$

$$52 f(x) = x^{2/3}; \quad a = -8$$

Exer. 53–56: Let n denote an arbitrary integer. Sketch the graph of f and find $\lim_{x \rightarrow n^-} f(x)$ and $\lim_{x \rightarrow n^+} f(x)$.

$$53 f(x) = (-1)^n \text{ if } n \leq x < n+1$$

$$54 f(x) = n \text{ if } n \leq x < n+1$$

$$55 f(x) = \begin{cases} x & \text{if } x = n \\ 0 & \text{if } x \neq n \end{cases} \quad 56 f(x) = \begin{cases} 0 & \text{if } x = n \\ 1 & \text{if } x \neq n \end{cases}$$

Exer. 57–60: Let $\llbracket \cdot \rrbracket$ denote the greatest integer function and n an arbitrary integer. Find

$$(a) \lim_{x \rightarrow n^-} f(x) \quad (b) \lim_{x \rightarrow n^+} f(x)$$

$$57 f(x) = \llbracket x \rrbracket \quad 58 f(x) = x - \llbracket x \rrbracket$$

$$59 f(x) = -\llbracket -x \rrbracket \quad 60 f(x) = \llbracket x \rrbracket - x^2$$

Exer. 61–64: Use the sandwich theorem to verify the limit.

$$61 \lim_{x \rightarrow 0} (x^2 + 1) = 1 \quad (\text{Hint: Use } \lim_{x \rightarrow 0} (|x| + 1) = 1.)$$